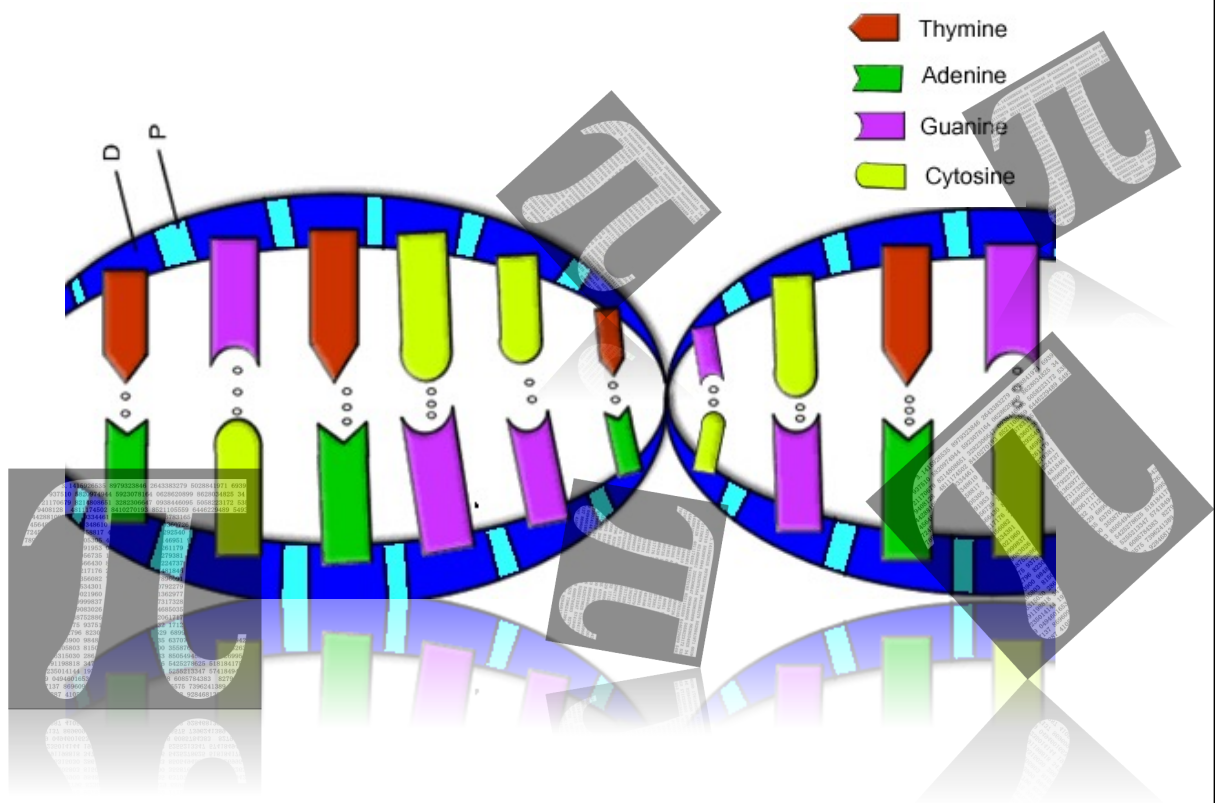


Computational Intelligence II: Evolutionary Optimisation

Chapter 4: Theoretical analysis of Genetic Algorithms



Schemata

⌘ Most of the theoretical analyses of GAs rely on binary alphabets and the notion of a **schema**:

- ⊙ A template providing the means of exploring similarities between chromosomes and analysing the effects of the genetic operators.

⌘ A schema is formed by the representation alphabet $A = \{0,1\}$ and the special symbol '*' standing for 'don't-care'.

Example:

The schema '11*01*10' of length $m=8$ matches all bit-strings:

'11001010',
'11001110',
'11101010',
'11101110',

⌘ In this way, each string can be matched by 2^m schemata and a GA population of n members of m length each, contains between 2^m and $n \cdot 2^m$ different schemata.

⌘ A schema has geometric interpretation based on an m -dimensional hypercube. Within this, every schema is a hyperplane or subset of the search space.

- ⊙ For example, for $m=3$, '***' gives the whole space, '001' one corner of the cube, '11*' a line and '**0' a plane.

⌘ The **order** $o(\cdot)$ of a schema is defined as the number of specific bits it contains.

⊙ For example, $o(11*01*10) = 8 - 2 = 6$.

⊙ A schema S defines a hyperplane of dimensionality given by $\text{length}(m) - o(S)$.

⌘ Another useful concept is that of the **defining length** $\delta(\cdot)$. This is the distance between the first and the last specific bits.

Examples:

$$\delta(11*01*10) = 7$$

$$\delta(11*01*) = 4$$

$$\delta(***1**) = 0$$

Reproductive schema growth

- ⌘ We can define as $\xi(\mathbf{S}, t)$ the **number of strings** in a population at generation t , matched by the schema $\mathbf{S}$.
- ⌘ We also define a **fitness** $f(\mathbf{S})$ equal to the average fitness of all strings in the population matched by $\mathbf{S}$.
- ⌘ Then, after n fitness-proportionate selections with replacement from a non-overlapping population of size n , the number of strings matched by $\mathbf{S}$ in the next generation is:

$$\xi(\mathbf{S}, t+1) = n \cdot \xi(\mathbf{S}, t) \cdot \frac{f(\mathbf{S})}{\sum_{k=1}^n f_k}$$

- ⌘ Using f_{avg} to denote average population fitness, this relation becomes:

$$\xi(\mathbf{S}, t+1) = \xi(\mathbf{S}, t) \cdot \frac{f(\mathbf{S})}{f_{\text{avg}}}$$

- ⌘ This means that growth and decay of schemata in the population depend on the ratio of their fitness over the average population fitness. The above is referred as to the **reproductive schema growth equation (RSGE)**.

Exponential sampling

- ⌘ We now assume that the fitness of a schema stays above average by a percentage ε , that is:

$$f(S) = (1 + \varepsilon) \cdot f_{\text{avg}}$$

- ⌘ Then, the RSGE becomes:

$$\xi(S, t+1) = \xi(S, t) \cdot (1 + \varepsilon)$$

- ⌘ Starting from generation $t=0$, we obtain the geometric progression version of RSGE:

$$\xi(S, t) = \xi(S, 0) \cdot (1 + \varepsilon)^t$$

- ⌘ The formula implies that above-average schemata receive an exponential number of string samples in the future generations.

- ⊙ Nevertheless, this exponential increase is achieved by the reproduction phase alone and it is ineffective, because no new regions of the search space are introduced to the population.
- ⊙ This is the responsibility of the other two genetic operators.
- ⊙ Assume that two selected members are subjected to 1-point crossover with probability p_c or are simply reproduced to the new population with $1-p_c$.
- ⊙ Mutation is applied to all generated offspring and flips each gene with a low probability p_m .

- ⌘ Since a crossover site in an m -bit chromosome is selected among $m-1$ possible break-sites, the probability of disrupting a schema S is proportional to $\delta(S) / (m-1)$.
- ⌘ Also, since such disruption may occur only when crossover takes place, the quantity $1 - p_c \cdot \delta(S) / (m-1)$ is a lower bound of the probability of survival of S . Thus we get:

$$\xi(S, t+1) \geq \xi(S, t) \cdot \frac{f(S)}{f_{\text{avg}}} \cdot \left(1 - p_c \cdot \frac{\delta(S)}{m-1} \right)$$
- ⌘ The above states that above-average, short (of small defining length) schemata will be sampled at exponentially increasing rates.
- ⌘ To incorporate mutation, we use the observation that in order for S to stay undisrupted, all its specific bits must be unaffected by mutation. Since a single allele survives with independent probability $1-p_m$, S survives with probability $(1-p_m)^{o(S)}$. Because $p_m \ll 1$, this quantity is approximated by $1 - o(S) \cdot p_m$. Thus, we get:

$$\xi(S, t+1) \geq \xi(S, t) \cdot \frac{f(S)}{f_{\text{avg}}} \cdot \left(1 - p_c \cdot \frac{\delta(S)}{m-1} - o(S) \cdot p_m \right)$$

The Schema theorem

- ⌘ The last formula is referred to as the ***Schema Theorem or Fundamental Theorem of GAs***. It states that above-average, short, low-order schemata receive exponentially increasing trials in subsequent generations. It places a lower bound on the expected number of strings from a schema S .
- ⌘ This leads to the so-called ***Building Block Hypothesis***, which states that a GA seeks near-optimal performance through the juxtaposition of short, low-order, high-performance schemata, called ***building blocks***.
- ⌘ Thus, instead of searching for optimal solutions by performing every conceivable combination, the GAs construct progressively better strings from the best partial solutions of the past samplings.
- ⌘ Although not all schemata are processed in a useful manner (that is without being disrupted from crossover and mutation) it can be shown that at least n^3 of them are processed usefully. This is related to the ***Implicit Parallelism of GAs***.

<END of Chapter 4>